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Review

Distribution patterns of narcotic drugs, psychotropic substances, and new psychoactive substances among reported drug users in china (2014–2026): A systematic review and Meta-Analysis

Qianning Wang [a,1], Zhen Qin [b,1], Lang Qin [a], Xing Xing [c], Zekun Wang [b], Haoliang Cui [a], Jianyi Zhang [a], Yiwen Xing [a], Jingyuan Chen [a], Mingchen Zhao [b,d], Wenkai Luo [a], He Zhu [c,*], Zhongwei Jia [a,**]

[a] Department of Global Health, School of Public Health, Peking University, Beijing 100191, China
[b] Department of Epidemiology and Biostatistics, School of Public Health, Peking University, Beijing 100191, China
[c] China Center for Health Development Studies, Peking University, Beijing 100191, China
[d] Beijing Office of Global Health, Beijing Center for Disease Prevention and Control, Beijing 100013, China

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ABSTRACT

Purpose: This study aims to examine the distribution of narcotic drugs, psychotropic substances, and new psychoactive substances (NPS) among reported drug users in China.
Methods: This study searched peer-reviewed literature databases (i.e., CNKI and WANFANG for studies in Chinese, and Web of Science, PubMed, and PsycINFO for studies in English) for studies reporting drug use in China published between 2014 and 2026. Meta-regression was used to estimate pooled proportions of drug use, stratified by pandemic period, age group, and geographic region.
Results: The pooled proportions of narcotic drugs and psychotropic substances among reported drug users were estimated to be 6.87% and 86.32%; 5.50% and 91.46% in 2014–2019, 7.96% and 71.62% in 2020–2026; 9.42% and 71.19% among adolescents, 6.50% and 83.81% among adults; 7.08% and 78.35% in eastern areas, 4.84% and 89.99% in central areas, 7.20% and 85.21% in western areas. NPS accounted for less than 0.001%, with similarly low proportions in all subgroups. Significant geographic differences were observed for all substance types ($P < 0.05$), differences across population groups were observed for NPS ($P < 0.001$), and no statistically significant differences were detected across pandemic periods ($P > 0.05$).
Conclusion: Psychotropic substances accounted for the largest proportion, underscoring the need to prioritize their prevention and control. Geographic variations highlight the importance for region-specific strategies, targeting narcotic drugs in western areas, psychotropic substances and NPS in central areas. Population-specific interventions may also be warranted, particularly for adolescents, who exhibit higher proportions of NPS use.

1. Introduction

The global drug use problem is becoming increasingly severe, posing significant threats to public health and the economy. According to the United Nations Office on Drugs and Crime (UNODC), the number of drug users worldwide has risen by 20% over the past decade (United Nations Office on Drugs and Crime, 2024). This increase has resulted in approximately 600,000 deaths annually due to psychoactive substance use (World Health Organization, 2024). Moreover, for each drug user, the potential reduction in criminal activity and the associated savings in criminal justice costs have been estimated to reach as high as $193,440, with even the lower bound of such savings amounting to $621 (Fardone et al., 2023). Given these trends and their societal impact, understanding drug use patterns is essential for identifying high-risk groups, monitoring changes over time, and informing targeted prevention and intervention strategies.

* Corresponding author at: China Center for Health Development Studies, Peking University, Beijing 100191, China.
** Corresponding author at: Department of Global Health, School of Public Health, Peking University, Beijing 100191, China.
E-mail addresses: he.zhu@pku.edu.cn (H. Zhu), urchinjj@163.com (Z. Jia).
[1] Joint first authors.

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Globally, several epidemiological research has examined patterns of drug use across diverse populations and geographic regions, using large-scale surveys, surveillance systems, and systematic reviews (Degenhardt et al., 2008, 2019; Organisation for Economic Co-operation and Development, 2025; Shen et al., 2023). In recent years, however, pandemic-related restrictions during COVID-19, such as border closures, social distancing, and the shutdown of recreational settings, disrupted traditional drug supply chains and consumption environments, which may lead to shortages of classic street drugs and shifts toward alternative substances (Trana & Maida, 2021). These changes may have affected drug availability and drug use patterns, underscoring the importance of examining of the post-pandemic period to provide context-specific insights for informing prevention and control policies.

In contrast, research on drug use in China remains not sufficient potentially due to cultural and legal factors, as well as challenges related to data accessibility. Existing studies have primarily focused on surveys assessing drug awareness and usage among specific populations, particularly students and individuals in compulsory drug treatment programs (Xiong et al., 2022; Zhu & Chen, 2018; Zou et al., 2017). Several systematic reviews have reported significant disparities in the prevalence of illicit drug use across genders and regions among students (Jia et al., 2018; Zhu et al., 2021). Notably, we previously conducted a systematic review spanning the period from 2003 to 2013, synthesizing available evidence on the overall prevalence of student drug use in mainland China (Jia et al., 2018). While this earlier study provided valuable insights into specific subgroups, it was largely limited to particular age cohorts and settings. To our knowledge, no studies have systematically examined changes in drug use patterns in China before and after the COVID-19 pandemic, leaving a critical gap in understanding how the pandemic may have affected drug use patterns within China's unique social and regulatory context.

In China, several controlled substances are commonly recognized, including narcotic drugs, psychotropic substances, and new psychoactive substances (NPS), providing a framework for a better understanding of drug use patterns. Narcotic drugs refer to substances often derived from natural sources and regulated under the 1961 Single Convention on Narcotic Drugs (United Nations Office on Drugs and Crime, 1961). Psychotropic substances involve synthetic or semi-synthetic compounds that act on the central nervous system to alter mental states and are regulated under the 1971 Convention on Psychotropic Substances (United Nations Office on Drugs and Crime, 1971). NPS refer to newly developed or unregulated synthetic compounds not explicitly scheduled under the 1961 or 1971 Conventions but pose risks due to their psychoactive properties (United Nations Office on Drugs and Crime, 2025). These controlled substances were found to be more prevalent worldwide, especially after the pandemic (Trana & Maida, 2021).

This study aims to synthesize evidence on the distribution patterns of narcotic drugs, psychotropic substances, and NPS among reported drug users in China. It also seeks to examine variations across population groups, geographic regions, and pre- versus post-COVID-19 periods. Although nationally representative prevalence data are limited, a sufficient number of primary studies exist to enable a systematic review and *meta*-analysis of these outcomes. This approach could provide a nuanced understanding of substance use patterns and context-specific evidence to inform targeted prevention and control strategies in the post-pandemic era.

2. Methods

We conducted a systematic review of peer-reviewed studies in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) (Page et al., 2021). The protocol was registered with the International Prospective Register of Systematic Reviews, known as PROSPERO, registration number CRD42024571456 (National Institute for Health and Care Research, 2011).

2.1. Literature search strategy

We adopted a database selection strategy consistent with our previous study, expanding the search scope to capture a broader range of relevant literature. Specifically, we searched CNKI and WANFANG for Chinese-language studies, and Web of Science, PubMed, and PsycINFO for English-language studies. The search strategy was developed in consultation with a specialist drug librarian and applied using a comprehensive set of predefined search terms (Supplementary Table S1). At the database-search level, filters were applied to limit studies to those published between January 1, 2014, and March 1, 2026, and to exclude non-original publications (e.g., reviews, editorials, commentaries) as well as studies published in languages other than Chinese or English. The year 2014 was selected as the starting point to ensure consistency in substance classification, surveillance reporting standards, and comparability of proportional distribution outcomes, as earlier studies were not systematically designed to report detailed substance-type distributions required for the present analytic framework.

2.2. Screening and quality assessment

We created an EndNote (version X.8) library to catalog studies and remove duplicates. Title and abstract screening were conducted independently by two authors (QW and ZQ). At this stage, studies were excluded if they focused on countries other than China, were non-population-based (e.g., chemical or animal experiments), or did not report a specific geographic region or drug type. Full-text assessment was also performed independently by the same two authors. Studies were excluded if they did not report, or allow calculation of, the relative distribution (i.e., proportions) of different types of substances among individuals who reported drug use in China, or if full-text access could not be obtained after attempts to contact the corresponding authors. Any disagreements at either stage were resolved through discussion with a third author (HZ).

The risk of bias in the remaining studies was assessed using the Joanna Briggs Institute (JBI) Checklist (Joanna Briggs Institute, 2020). The checklist includes: sample frame appropriate to address the target population; study participants sampled in an appropriate way; sample size adequate; study subjects and the setting described in detail; data analysis conducted with sufficient coverage of the identified sample; valid methods used for the identification of the condition; condition measured in a standard, reliable way for all participants; appropriate statistical analysis; response rate adequate, and if not, low response rate managed appropriately. Two authors (QW and ZQ) independently rated the risk scores, classifying studies as high-risk (scores 0–3), moderate-risk (scores 4–6), or low-risk (scores 7–9). Any discrepancies were resolved through discussion with a third author (HZ). Overall, two studies were classified as high risk and excluded, while 52 studies with moderate or low risk of bias were included for data extraction (Fig. 1).

2.3. Data extraction

Data from eligible studies were extracted into a predesigned, purpose-built database using Microsoft Excel 2019 by two authors (QW and ZQ). The extracted data included the first author, publication year, data year, study design, study province, age group, population group, and sample characteristics (e.g., male-to-female ratio, sample size, types of drug use, and their corresponding samples). To ensure the accuracy of data extraction, all studies published in both Chinese and English were reviewed by bilingual authors. When further clarification was needed, we contacted all the authors to discuss it.

Missing and zero values were handled as follows: missing data were excluded from the analysis, and zero values were carefully reviewed to determine whether they were valid or resulted from reporting errors. Any discrepancies were resolved through discussion with a third author

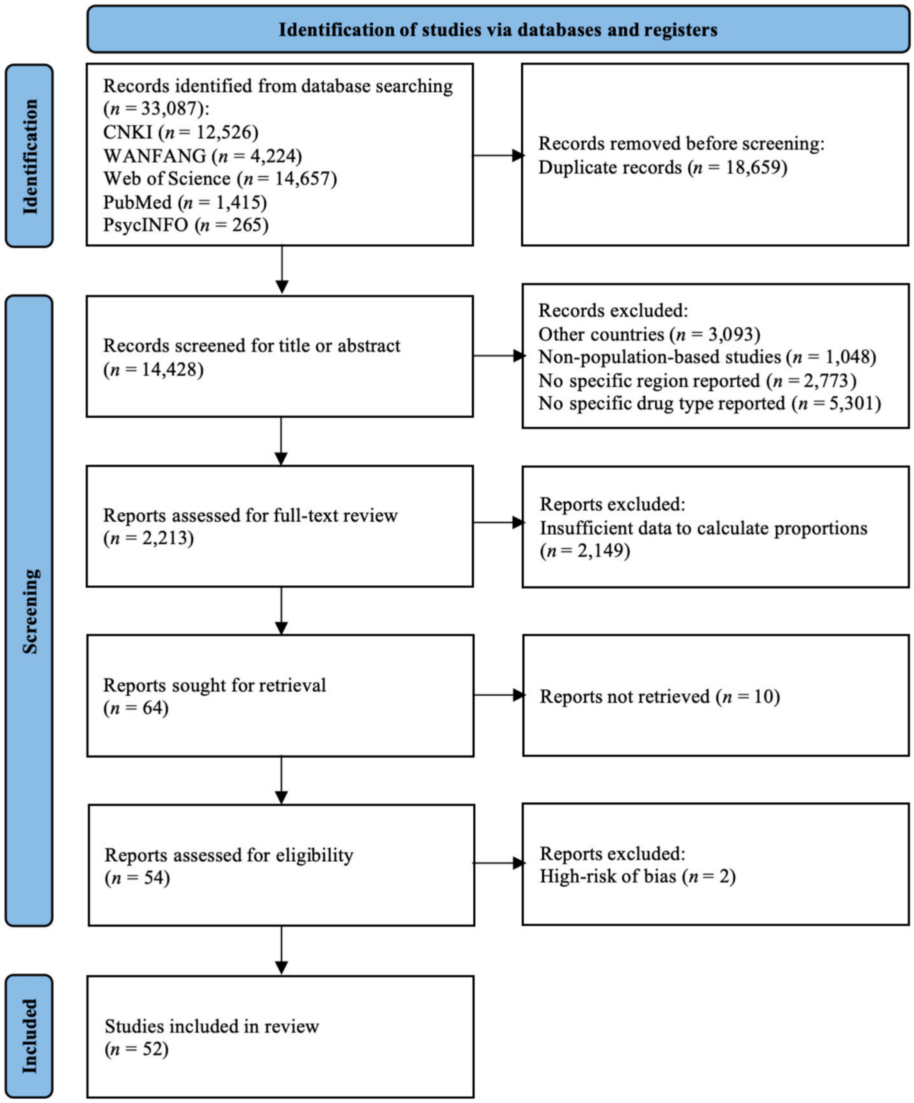

Fig. 1. PRISMA flow diagram for the systematic review.

(HZ).

2.4. Statistical analysis

We first described the distribution of narcotic drugs, psychotropic substances, and NPS among drug users in the included studies. We then stratified the data by pandemic period (period 1: 2014–2019 and period 2: 2020–2026), age group (i.e., adolescents aged 10–19 years and adults aged 20 years and above), population type (i.e., prison populations, compulsory drug treatment populations, treatment-seeking populations, sentinel surveillance populations, students, and the general population), and geographic region (i.e., eastern areas: Beijing, Fujian, Guangdong, Jiangsu, Macao, Shandong, Shanghai, and Zhejiang; central areas: Anhui, Henan, Hubei, Hunan, and Jiangxi; western areas: Chongqing, Gansu, Guangxi, Guizhou, Inner Mongolia, Ningxia, Qinghai, Shaanxi, Shanxi, Sichuan, and Yunnan).

The double arcsine transformation was applied to convert the original proportions, ensuring that the transformed data met normality assumptions and approximated a normal distribution (Moreno et al., 2012). Both pooled and subgroup analyses were performed using *meta*-regression, which accounted for study-specific weights and variances and applied a random-effects model (Moreno et al., 2012). Heterogeneity among the included studies was assessed using I^2 and τ^2 statistics (Higgins & Sally, 2008). Sensitivity analyses were conducted to evaluate the robustness of the results by excluding studies that significantly impacted the pooled outcomes and to identify potential sources of heterogeneity (Higgins & Sally, 2008). Publication bias was examined using Begg's test and funnel plots (Higgins & Sally, 2008). All statistical analyses were conducted in R version 4.3.0. Two-tailed significance tests used a cut-off value of $P = 0.05$.

3. Results

3.1. Study characteristics

A total of 52 eligible studies, comprising 83 data records, were included in the *meta*-analysis, covering 112,091 samples from 24 provinces in China (Table 1). All included studies adopted a cross-sectional design and reported lifetime substance use. The records included 47 from 2014 to 2019 and 36 from 2020 to 2026. The samples were predominantly male, with a mean male-to-female ratio of 5.10, and mostly adults (approximately 90.36%), with the remainder being adolescents. The specific population types included prison populations, compulsory drug treatment populations, treatment-seeking populations, sentinel surveillance populations, students, and the general population.

Table 1
Characteristics of the eligible studies.

First author	Publication year	Data year	Study province	Age group	Population group	Male-to-female ratio	Sample size	Narcotic drugs	Psychotropic substances	NPS
Chen, B.	2016	2014	Anhui	Adults	Compulsory drug treatment populations	3.63	987	883	104	0
Jian, Y. Y.	2023	2017	Anhui	Adults	Sentinel surveillance populations	9.49	472	16	456	0
Jian, Y. Y.	2023	2018	Anhui	Adults	Sentinel surveillance populations	6.46	428	14	413	1
Jian, Y. Y.	2023	2019	Anhui	Adults	Sentinel surveillance populations	4.23	251	24	222	5
Jian, Y. Y.	2023	2020	Anhui	Adults	Sentinel surveillance populations	6.27	112	6	104	2
Jian, Y. Y.	2023	2021	Anhui	Adults	Sentinel surveillance populations	4.44	121	13	94	14
Li, G. Y.	2025	2017	Beijing	Adults	Prison populations	NA	2,923	681	2,242	0
Li, G. Y.	2025	2018	Beijing	Adults	Prison populations	NA	2,761	696	2,065	0
Li, G. Y.	2025	2019	Beijing	Adults	Prison populations	NA	2,336	724	1,612	0
Li, G. Y.	2025	2020	Beijing	Adults	Prison populations	NA	970	404	566	0
Li, G. Y.	2025	2021	Beijing	Adults	Prison populations	NA	1,120	482	638	0
Mao, Z. T.	2022	2021	Beijing	Adults	Compulsory drug treatment populations	5.08	414	101	313	0
Mao, Z. T.	2023	2022	Beijing	Adults	Compulsory drug treatment populations	3.29	304	79	225	0
Zhou, H. T.	2019	2018	Chongqing	Adults	Compulsory drug treatment populations	2.00	80	0	80	0
Xue, Q.	2023	2021	Chongqing	Adults	Compulsory drug treatment populations	1.01	228	84	144	0
Ding, Z. M.	2022	2021	Fujian	Adults	Sentinel surveillance populations	4.20	5,400	974	4,131	295
Zou, X. Y.	2025	2025	Fujian	Adolescents	Students	NA	383	13	157	75
Du, X. Y.	2021	2019	Gansu	Adults	Compulsory drug treatment populations	NA	102	3	36	63
Lu, Y. M.	2016	2015	Guangdong	Adults	Compulsory drug treatment populations	10.59	313	79	234	0
Wu, J. R.	2020	2016	Guangdong	Adults	Prison populations, Compulsory drug treatment populations, Treatment-seeking populations	9.74	395	0	395	0
Wu, J. R.	2020	2017	Guangdong	Adults	Prison populations, Compulsory drug treatment populations, Treatment-seeking populations	12.47	1,230	0	1,230	0
Wu, J. R.	2020	2018	Guangdong	Adults	Prison populations, Compulsory drug treatment populations, Treatment-seeking populations	10.74	759	0	759	0
Huang, X. X.	2020	2018	Guangdong	Adults	Treatment-seeking populations	7.23	39	39	0	0
Peng, X. X.	2019	2018	Guangdong	Adults	Compulsory drug treatment populations, Treatment-seeking populations	6.74	744	263	481	0
Chen, X.	2025	2023	Guangdong	Adults	Sentinel surveillance populations	9.16	13,326	7,832	4,092	0
Zhang, L.	2015	2014	Guangxi	Adults	Prison populations	3.22	1,448	732	540	176
Chen, Y. H.	2025	2021	Guangxi	Adults	Sentinel surveillance populations	17.79	263	181	65	0
Chen, Y. H.	2025	2022	Guangxi	Adults	Sentinel surveillance populations	39.80	204	148	40	0
Chen, Y. H.	2025	2023	Guangxi	Adults	Sentinel surveillance populations	13.55	160	90	64	0
Yu, Y.	2025	2022	Guangxi	Adults	Treatment-seeking populations	1.00	384	63	0	0
Zhang, D. R.	2017	2015	Guizhou	Adults	Compulsory drug treatment populations	1.40	420	22	398	0
Wang, Z. Q.	2019	2017	Guizhou	Adults	Compulsory drug treatment populations	9.79	16,177	10,588	5,589	0
Guo, H.	2018	2016	Henan	Adults	Sentinel surveillance populations	0.74	3,497	0	3,493	4
Wang, Z. W.	2024	2021	Henan	Adults	Treatment-seeking populations	1.13	65	19	46	0
Li, S. Q.	2017	2015	Hubei	Adults	Compulsory drug treatment populations	8.72	335	0	335	0
Tang, J.	2015	2014	Hunan	Adults	Compulsory drug treatment populations	6.19	2,146	1,012	1,134	0
Du, X. Y.	2021	2019	Inner Mongolia	Adults	Compulsory drug treatment populations	NA	121	2	42	77

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Table 1 (*continued*)

First author	Publication year	Data year	Study province	Age group	Population group	Male-to-female ratio	Sample size	Narcotic drugs	Psychotropic substances	NPS
Cao, Q. L.	2021	2017	Jiangsu	Adults	Compulsory drug treatment populations	3.50	502	294	208	0
Zhu, D. Y.	2018	2017	Jiangsu	Adults	Compulsory drug treatment populations	NA	1,223	108	1,115	0
Yang, H. Y.	2024	2021	Jiangsu	Adults	Sentinel surveillance populations	NA	400	25	355	0
Yang, H. Y.	2024	2022	Jiangsu	Adults	Sentinel surveillance populations	NA	214	8	195	0
Yang, H. Y.	2024	2023	Jiangsu	Adults	Sentinel surveillance populations	NA	311	22	252	0
Xiao, H.	2018	2017	Jiangxi	Adolescents	Compulsory drug treatment populations	5.27	496	64	432	0
Tang, L. J.	2019	2014	Macao	Adolescents	Sentinel surveillance populations	2.55	45	7	37	1
Tang, L. J.	2019	2015	Macao	Adolescents	Sentinel surveillance populations	1.00	69	13	55	1
Tang, L. J.	2019	2016	Macao	Adolescents	Sentinel surveillance populations	1.27	30	6	24	0
Tang, L. J.	2019	2017	Macao	Adolescents	Sentinel surveillance populations	1.10	23	7	16	0
Xiong, R. S.	2022	2018	Macao	Adolescents	Students	NA	638	0	361	277
Du, X. Y.	2021	2019	Ningxia	Adults	Compulsory drug treatment populations	NA	132	36	55	41
Kuang, J. Y.	2025	2018	Qinghai	Adults	Sentinel surveillance populations	NA	1,188	748	354	0
Kuang, J. Y.	2025	2019	Qinghai	Adults	Sentinel surveillance populations	NA	1,205	666	500	0
Kuang, J. Y.	2025	2020	Qinghai	Adults	Sentinel surveillance populations	NA	1,045	604	363	0
Kuang, J. Y.	2025	2021	Qinghai	Adults	Sentinel surveillance populations	NA	1,225	507	637	0
Kuang, J. Y.	2025	2022	Qinghai	Adults	Sentinel surveillance populations	NA	1,200	749	379	0
Huang, X. D.	2020	2019	Shaanxi	Adults	Compulsory drug treatment populations	21.54	1,546	1,315	231	0
Li, H. B.	2025	2020	Shaanxi	Adults	Compulsory drug treatment populations	NA	2,000	0	267	0
Yu, Y.	2025	2021	Shandong	Adults	Treatment-seeking populations	1	453	0	425	0
Guo, M. Z.	2025	2023	Shandong	Adults	Treatment-seeking populations	NA	3,235	0	1,761	0
Xia, Y. W.	2015	2014	Shanghai	Adults	Compulsory drug treatment populations	0.93	822	312	510	0
Fu, J.	2018	2017	Shanghai	Adults	Treatment-seeking populations	3.68	2,514	1,946	568	0
Chen, J. S.	2025	2023	Shanghai	Adults	Sentinel surveillance populations	3.93	2,801	0	1,811	0
Yuan, B.	2015	2014	Sichuan	Adults	Compulsory drug treatment populations	0.43	447	352	95	0
Chen, H.	2018	2015	Sichuan	Adults	Compulsory drug treatment populations, Treatment-seeking populations, Sentinel surveillance populations	2.55	2,718	2,484	234	0
Wang, P.	2021	2020	Sichuan	Adults	Compulsory drug treatment populations	1.27	419	95	324	0
Li, Y. Q.	2025	2024	Sichuan	Adults	Compulsory drug treatment populations	1.05	1,068	154	833	0
Sun, C. Y.	2026	2016	Sichuan	Adults	Sentinel surveillance populations	0.95	294	0	294	0
Sun, C. Y.	2026	2017	Sichuan	Adults	Sentinel surveillance populations	0.84	316	0	316	0
Sun, C. Y.	2026	2018	Sichuan	Adults	Sentinel surveillance populations	0.79	330	0	330	0
Sun, C. Y.	2026	2019	Sichuan	Adults	Sentinel surveillance populations	0.79	320	0	320	0
Sun, C. Y.	2026	2020	Sichuan	Adults	Sentinel surveillance populations	0.77	110	0	110	0
Sun, C. Y.	2026	2021	Sichuan	Adults	Sentinel surveillance populations	0.81	421	0	421	0
Sun, C. Y.	2026	2022	Sichuan	Adults	Sentinel surveillance populations	0.68	103	0	103	0
Sun, C. Y.	2026	2023	Sichuan	Adults	Sentinel surveillance populations	0.78	468	0	468	0

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Table 1 (*continued*)

First author	Publication year	Data year	Study province	Age group	Population group	Male-to-female ratio	Sample size	Narcotic drugs	Psychotropic substances	NPS
Sun, C. Y.	2026	2024	Sichuan	Adults	Sentinel surveillance populations	0.77	382	0	382	0
Su, S.	2018	2015	Yunnan	Adults	Compulsory drug treatment populations	9.34	327	295	32	0
Su, S.	2018	2017	Yunnan	Adults	Compulsory drug treatment populations	5.15	2,081	765	1,316	0
Mo, G. Y.	2016	2015	Yunnan	Adolescents	Compulsory drug treatment populations, Sentinel surveillance populations	4.36	1,585	588	997	0
Kang, W. L.	2017	2017	Yunnan	Adults	Compulsory drug treatment populations, Treatment-seeking populations, Sentinel surveillance populations	1.02	11,478	7,166	4,312	0
Gu, Y.	2020	2020	Yunnan	Adults	General populations	1.34	64	0	64	0
Zhang, L. Y.	2024	2022	Yunnan	Adults	Sentinel surveillance populations	NA	2,529	1,925	508	0
Li, X. M.	2026	2024	Yunnan	Adults	Compulsory drug treatment populations	NA	1,021	357	347	0
Shen, H. L.	2016	2014	Zhejiang	Adults	Compulsory drug treatment populations	3.89	460	54	406	0
Jia, D.	2024	2020	Zhejiang	Adults	Compulsory drug treatment populations	2.61	415	182	233	0

Note: The study design of the included research was cross-sectional, and the study period focused on lifetime use. NPS: new psychoactive substance. NA: not available.

3.2. *Estimation of narcotic drug use, stratified by subgroups*

The pooled proportion of lifetime narcotic drug use among reported drug users in the included studies was estimated to be 6.87% (95% CI: 3.10%, 14.55%) (Fig. 2A). In subgroup analyses (Table 2), the pooled proportions of narcotic drug use did not differ significantly between 2014–2019 (5.50%) and 2020–2026 (7.96%) ($P = 0.664$), nor between adolescents (9.42%) and adults (6.50%) ($P = 0.627$). By population type, the highest pooled proportion was observed in prison populations (34.42%), but no statistically significant between-group difference was detected ($P = 0.162$). Pooled proportions differed significantly across geographic regions (eastern: 7.08%, central: 4.84%, and western: 7.20%; $P < 0.001$). At the provincial level, the proportions of narcotic drug use varied widely ($P < 0.001$), with the highest proportion observed in Shanxi (85.06%).

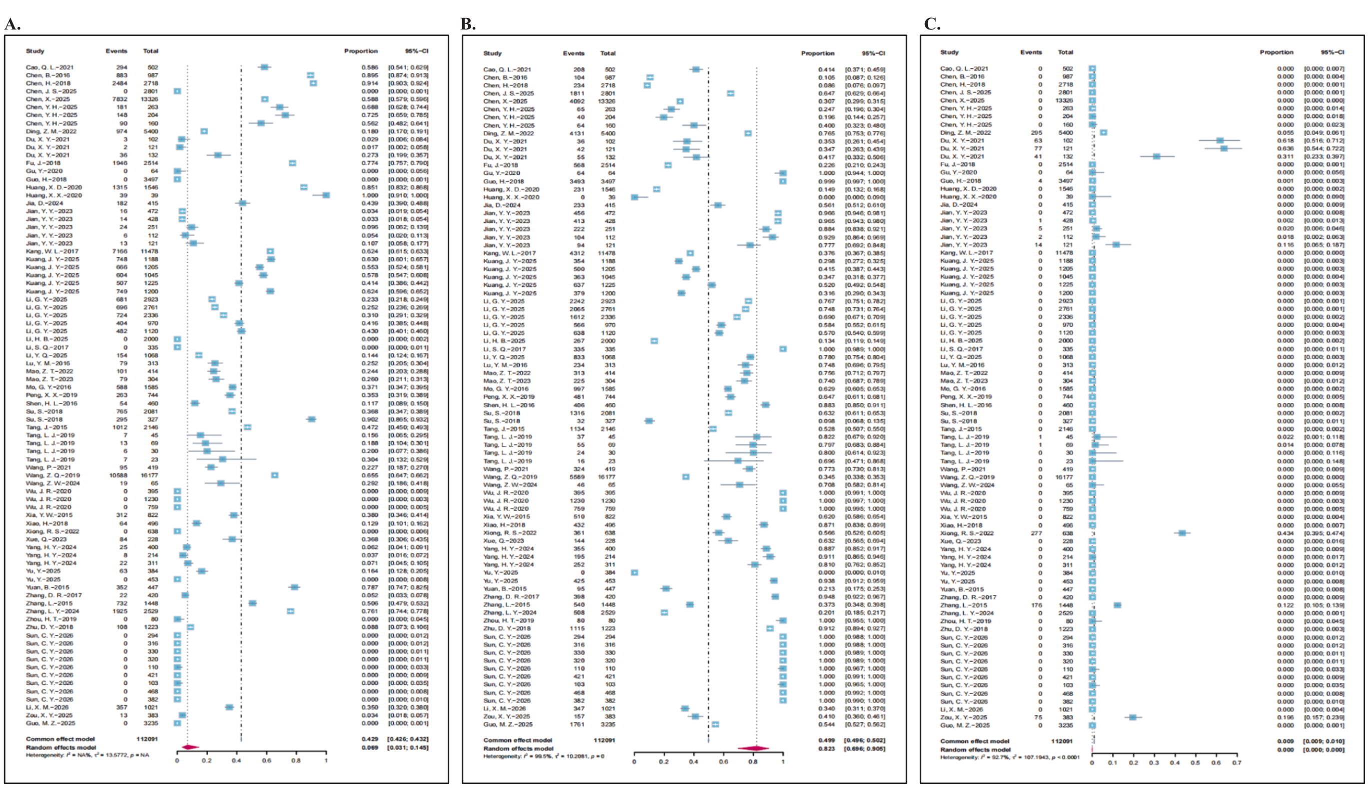

Fig. 2. The estimated pooled proportion of narcotic drugs, psychotropic substances, and NPS among reported drug users in the included studies.

Table 2
The estimated pooled proportion of drug use by period, age group, population group, geographic region, and province.

	Number of data records	Narcotic drugs					Psychotropic substances					NPS				
		Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2	Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2	Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2
Period																
2014–2019	37	28,109	5.50 (1.27, 20.89)	0.664	99.31	19.29	27,619	91.46 (74.08, 97.57)	0.061	99.34	15.20	460	0.00 (0.00, 0.35)	0.319	90.34	45.27
2020–2026	46	20,012	7.96 (3.23, 18.30)		99.25	9.89	28,276	71.62 (54.63, 84.10)		99.52	6.27	572	0.00 (0.00, 0.00)		92.70	154.79
Age group																
Adolescents	8	698	9.42 (2.67, 28.27)	0.627	96.28	3.44	2,079	71.19 (59.13, 80.84)	0.136	96.64	0.52	354	0.38 (0.02, 8.80)	0.001	91.27	13.26
Adults	75	47,423	6.50 (2.67, 14.98)		99.48	15.02	53,816	83.81 (69.70, 92.10)		99.51	11.91	678	0.00 (0.00, 0.00)		89.13	131.83
Population group																
Prison populations	7	3,738	34.42 (27.54, 42.03)	0.162	98.74	0.18	7,709	63.98 (53.91, 72.95)	0.107	99.24	0.31	176	0.00 (0.00, 99.90)	<0.001	0.00	182.78
Compulsory drug treatment populations	27	17,236	19.63 (8.70, 38.50)		99.45	5.97	15,038	63.30 (45.59, 78.02)		99.48	3.57	181	0.00 (0.00, 0.16)		19.50	304.52
Treatment-seeking populations	5	2,048	1.07 (0.00, 99.97)		98.96	118.97	2,754	5.19 (0.05, 85.22)		99.50	22.35	0	–		–	–
Sentinel surveillance populations	34	14,585	2.99 (0.75, 11.23)		99.19	14.91	21,404	93.88 (81.83, 98.12)		99.40	11.75	352	0.02 (0.00, 0.18)		98.26	12.64
Students	2	13	0.21 (0.00, 31.50)		0	9.17	518	48.93 (38.23, 59.72)		95.66	0.09	277	43.42 (39.62, 47.29)		–	–
General population	1	0	–		–	–	64	100.00 (0.00, 100.00)		–	–	0	–		–	–
Mixed population[a]	7	10,501	1.26 (0.01, 67.03)		99.55	35.53	8,408	98.74 (32.92, 99.99)		99.55	35.53	0	–		–	–
Geographic region																
Eastern areas	33	15,351	7.08 (6.95, 7.20)	<0.001	99.40	12.41	27,472	78.35 (63.30, 88.36)	0.001	99.56	4.56	649	0.00 (0.00, 0.03)	0.033	95.36	121.73
Central areas	12	2,053	4.84 (0.89, 22.38)		99.06	8.76	6,875	89.99 (66.92, 97.56)		99.17	6.71	103	0.12 (0.01, 1.68)		96.25	14.51
Western areas	38	30,717	7.20 (1.89, 23.75)		99.18	16.40	21,548	85.21 (56.92, 96.17)		99.12	18.93	280	0.00 (0.00, 0.05)		73.51	286.77
Province																
Anhui	6	956	12.05 (2.81, 39.35)	<0.001	99.48	3.70	1,393	85.06 (55.31, 96.32)	<0.001	99.47	3.58	22	0.39 (0.04, 3.75)	<0.001	81.12	6.05
Beijing	7	3,167	30.24 (24.95, 36.11)		97.60	0.12	7,661	69.76 (63.89, 75.05)		97.60	0.12	0	–		–	–
Chongqing	2	84	1.94 (0.00, 91.99)		0.00	13.73	224	98.06 (8.01, 100.00)		0.00	13.73	0	–		–	–
Fujian	2	987	8.27 (2.41, 24.71)		97.60	0.83	4,288	60.20 (34.14, 81.53)		99.50	0.59	370	10.49 (4.15, 24.05)		99.03	0.50
Gansu	1	3	2.94 (0.95, 8.72)		–	–	36	35.29 (26.65, 45.02)		–	–	63	61.76 (52.00, 70.66)		–	–

(*continued on next page*)

Table 2 (*continued*)

	Number of data records	Narcotic drugs Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2	Psychotropic substances Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2	NPS Number of samples	Proportion (%) (95% CI)	P value[b]	I^2 (%)	τ^2
Guangdong	7	8,213	0.80 (0.00, 98.76)		97.70	76.38	7,191	99.19 (1.00, 100.00)		98.83	78.86	0	–		–	–
Guangxi	5	1,214	52.02 (32.54, 70.91)		98.14	0.84	709	11.51 (1.39, 54.52)		89.22	5.90	176	0.00 (0.00, 99.98)		0.00	71.06
Guizhou	2	10,610	24.50 (2.69, 79.22)		99.61	3.14	5,987	75.50 (20.78, 97.31)		99.61	3.14	0	–		–	–
Henan	2	19	0.07 (0.00, 99.69)		0.00	47.36	3,539	97.95 (42.88, 99.97)		99.06	8.81	4	0.11 (0.04, 0.30)		0.00	0.00
Hubei	1	0	–		–	–	335	100.00 (0.00, 100.00)		–	–	0	–		–	–
Hunan	1	1,012	47.16 (45.05, 49.27)		–	–	1,134	52.84 (50.73, 54.95)		–	–	0	–		–	–
Inner Mongolia	1	2	1.65 (0.41, 6.37)		–	–	42	34.71 (26.78, 43.60)		–	–	77	63.64 (54.72, 71.71)		–	–
Jiangsu	5	457	10.89 (3.80, 27.42)		99.26	1.61	2,125	82.73 (66.14, 92.15)		99.15	1.02	0	–		–	–
Jiangxi	1	64	12.90 (10.23, 16.15)		–	–	432	87.10 (83.85, 89.77)		–	–	0	–		–	–
Macao	5	33	7.00 (0.80, 41.37)		0.00	5.89	493	73.39 (62.07, 82.30)		85.31	0.23	279	1.85 (0.14, 19.88)		85.29	5.10
Ningxia	1	36	27.27 (20.36, 35.49)		–	–	55	41.67 (33.57, 50.24)		–	–	41	31.06 (23.76, 39.45)		–	–
Qinghai	5	3,274	56.05 (49.08, 62.79)		97.29	0.10	2,233	37.65 (30.92, 44.89)		97.56	0.11	0	–		–	–
Shaanxi	1	0	–		–	–	267	13.35 (11.93, 14.91)		–	–	0	–		–	–
Shandong	2	0	–		–	–	2,186	80.89 (41.83, 96.14)		99.39	1.62	0	–		–	–
Shanghai	3	2,258	2.76 (0.00, 97.11)		99.50	33.64	2,889	48.86 (26.94, 71.21)		99.79	0.70	0	–		–	–
Shanxi	1	1,315	85.06 (83.19, 86.75)		–	–	231	14.94 (13.25, 16.81)		–	–	0	–		–	–
Sichuan	13	3,085	0.00 (0.00, 1.33)		99.29	108.41	4,230	100.00 (98.71, 100.00)		99.24	111.44	0	–		–	–
Yunnan	7	11,096	41.86 (13.62, 76.68)		99.56	3.98	7,576	52.81 (19.16, 84.08)		99.53	4.16	0	–		–	–
Zhejiang	2	236	24.41 (8.61, 52.54)		99.02	0.77	639	75.59 (47.46, 91.39)		99.02	0.77	0	–		–	–

Note: [a] Mixed population refers to studies including two or more population groups; [b] P value refers to the significance of the Q test in subgroup *meta*-analysis; cells with a dash (–) indicate that the subgroup contained only a single study, and the results are from the original study rather than pooled *meta*-analysis; therefore, measures of heterogeneity (I^2 and τ^2) are not applicable.

3.3. Estimation of psychotropic substance use, stratified by subgroups

The pooled proportion of lifetime psychotropic substance use among reported drug users in the included studies was estimated to be 82.32% (95% CI: 69.58%, 90.45%) (Fig. 2B). In subgroup analyses (Table 2), the pooled proportions of psychotropic substance use did not differ significantly between 2014–2019 (91.46%) and 2020–2026 (71.62%) (P = 0.061). No significant differences were observed between adolescents (71.19%) and adults (83.81%) (P = 0.136), nor across population types (P = 0.107). However, pooled proportions differed significantly across geographic regions (eastern: 78.35%, central: 89.99%, and western: 85.21%; P = 0.001). At the provincial level, psychotropic substance proportions varied widely (P < 0.001), with the highest proportion observed in Hubei and Sichuan (100.00%).

3.4. Estimation of NPS use, stratified by subgroups

The pooled proportion of lifetime NPS use among reported drug users in the included studies was estimated to be extremely low (< 0.001%; 95% CI: 4.14×10^{-11}%, 3.45×10^{-4}%) (Fig. 2C). In subgroup analyses (Table 2), the pooled proportions of NPS use did not differ significantly between 2014–2019 and 2020–2026 (P = 0.319). Significant differences were observed between adolescents and adults (0.38% vs. 0.00%; P = 0.001) and across population types (P < 0.001), with the highest pooled proportion observed among students (43.42%). Pooled proportions also varied significantly across geographic regions (P = 0.033). At the provincial level, NPS proportions showed considerable variation (P < 0.001), with the highest proportion observed in Inner Mongolia (63.64%).

3.5. Heterogeneity and publication bias

Heterogeneity was observed across studies for narcotic drugs (I^2 = 99.50%, τ^2 = 13.58), psychotropic substances (I^2 = 99.50%, τ^2 = 10.21), and NPS (I^2 = 93.90%, τ^2 = 107.19), with all Ps < 0.001. Sensitivity analyses indicated that no single study significantly influenced the pooled estimates. No evidence of publication bias was detected for narcotic drugs (P = 0.802), psychotropic substances (P = 0.936), or NPS (P = 0.053).

4. Discussion

This study examined the distribution of various types of substances, including narcotic drugs, psychotropic substances, and NPS, among reported drug users in China across different population groups, geographic regions, and pre- and post-COVID-19 periods. This analytical approach provides a framework for exploring both continuities and potential changes in substance use patterns in China, situating these patterns within a broader international context. The findings have important implications for risk stratification, targeted prevention, and the allocation of clinical and public health resources.

Consistent with previous research, psychotropic substances accounted for the largest proportion of reported drug use in China, followed by narcotic drugs and NPS. This pattern reflects long-standing characteristics of drug use patterns in China, where psychotropic substances such as methamphetamine and ecstasy have been widely reported and monitored (Chen et al., 2017; Yang et al., 2017). Recent data from the 2023 China Drug Situation Report indicate that quantities of psychotropic substances surged by 58.10% (China Drug Control Website, 2024). Narcotic drugs, including heroin and cannabis, also occupy a significant share of the market, which may be related to shifting public perceptions, particularly in the context of cannabis legalization in some countries and regions, where individuals may adopt a more open and tolerant attitude toward these substances (Farrelly et al., 2023; Guttmannova et al., 2016). One study in the U.S. found that following recreational cannabis legalization, 25% of the population held positive attitudes, with some former medical cannabis users partially shifting to recreational use (AminiLari et al., 2023). Although the proportion of NPS was relatively low, it is likely underreported due to rapid substance evolution, inconsistent classification, and limited detection capacity (Chen & Fu, 2020). This underreporting is further supported by wastewater-based analyses in multiple countries, which have reported lower detection frequencies for NPS relative to traditional drugs, suggesting methodological limitations rather than a true absence of use (Castiglioni et al., 2021).

Significant differences were observed across geographic regions for all types of substances; however, differences across population groups were observed only for NPS, and no statistically significant differences were observed between pre- and post-COVID-19 periods. Overall, psychotropic substances tended to account for a higher proportion in central and western regions, narcotic drugs showed substantial provincial variability, and NPS, although rare overall, exhibited localized concentrations in specific provinces. Geographic variation in drug use has been widely documented internationally and is primarily attributed to differences in access, pricing, trafficking routes, and enforcement intensity (Kerry & Saeid, 2020; National Institute of Justice, 2003). In China, provinces such as Guangdong, Yunnan, and Sichuan have reported higher levels of drug use, potentially reflecting their proximity to Southeast Asia, a region with established drug production and trafficking networks (Jia et al., 2018; Wang et al., 2024). Differences across population groups were confined to NPS, which may reflect distinct patterns of experimentation, diffusion, and social visibility compared with traditional drugs. This aligns with Golub et al.'s finding that drug use emerges from a dialectical relationship between prevailing subcultures and individual identity development, particularly among adolescents (Golub et al., 2005), suggesting that the social context and identity exploration of younger users may drive NPS experimentation. In contrast, international studies on cannabis use have reported age-related differences, highlighting that patterns of substance use across population groups may vary by drug type. For example, Martins et al. found that adults aged 26 and older reported greater perceived availability and a higher prevalence of cannabis use compared with younger adults or adolescents (P < 0.001) (Haug et al., 2017; Martins et al., 2016). No statistically significant differences were detected across periods for any of the three drug categories in China, suggesting that the distributional structure of drug types remained relatively stable during the pandemic. While prevalence rates may vary internationally across periods, there is currently no evidence confirming substantial changes in the distribution patterns of different drug categories (Trana & Maida, 2021). Moreover, statistical data on NPS remain limited and should be interpreted with caution.

These findings suggest that clinical screening and treatment efforts in China should continue to prioritize psychotropic substances (China Drug Control Website, 2024). Population-specific differences in NPS highlight the need for targeted prevention and improved clinical recognition. The observed geographic heterogeneity underscores the importance of region-specific prevention strategies, whereas the lack of significant differences across pandemic periods indicates a relative stability in substance use patterns (Jia et al., 2018; Wang et al., 2024). The stability supports sustained, rather than solely emergency-based, public health interventions. Although these findings were derived from China, the observed patterns illustrate broader general principles of substance use epidemiology: dominant substances drive overall clinical demand, emerging substances exhibit greater population heterogeneity, and short-term societal disruptions may have limited impact on underlying use structures. Given the limited data on NPS, efforts to improve surveillance systems, standardize definitions, and integrate different data sources, such as surveys, clinical records, forensic data, wastewater, could improve detection of emerging substances and support more timely, targeted prevention efforts.

Several limitations of this study should be acknowledged. First, sample coverage was uneven, with some population groups and provinces contributing relatively few data points. Second, heterogeneity was observed across studies, reflecting differences in data sources and reporting standards. Third, many studies were based on local samples rather than nationally representative data, restricting the generalizability of the findings. Fourth, the limited availability and potential underreporting of NPS data may further constrain the accuracy of the pooled estimates.

5. Conclusion

This study provides a systematic synthesis of evidence on the distribution patterns of narcotic drugs, psychotropic substances, and NPS among reported drug users in China. Overall, psychotropic substances accounted for the largest proportion. Geographical variations were observed for all substance types, whereas differences across population groups were observed only for NPS, and no differences related to the pandemic were identified. These findings highlight key principles of substance use epidemiology that can inform targeted prevention strategies, risk stratification, and health resource planning both within China and in comparable international contexts.

Data statement.

Data are available upon request from the corresponding author.

CRediT authorship contribution statement

Qianning Wang: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhen Qin:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lang Qin:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xing Xing:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zekun Wang:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoliang Cui:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jianyi Zhang:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yiwen Xing:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jingyuan Chen:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mingchen Zhao:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenkai Luo:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **He Zhu:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Data curation, Conceptualization. **Zhongwei Jia:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement and author contributions

QW, HZ, and ZJ designed the study. QW and ZQ independently searched and screened the literature using identical inclusion and exclusion criteria. QW and ZQ analyzed the data and wrote the first draft of the manuscript. QW and LQ interpreted the results and prepared the final manuscript. XX, ZW, HC, JZ, YX, JC, MZ, and WL revised the final manuscript. QW finalized the submitted manuscript. All authors have reviewed and approved the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.abrep.2026.100688.

Data availability

Data will be made available on request.

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Glossary

UNODC: United Nations Office on Drugs and Crime
NPS: New Psychoactive Substances
PRISMA: Preferred Reporting Items for Systematic reviews and Meta-Analyses
JBI: Joanna Briggs Institute